Intro to curves:

A line in $\mathbb{R}^n$ w/ pos P & dir v: $r(t) = P + tv$

Planes in $\mathbb{R}^3$ w/ pos P & normal vector n has plane equation: $n \cdot (x - P) = 0$

1. find equation of the plane through $(1,0,0)$, $(0,1,0)$ & $(0,0,1)$

$\vec{p} = \langle 1,0,0 \rangle$ ← Any position on the plane.

$\vec{n} = (j - i) \times (k - i)$

$= \begin{vmatrix} i & j & k \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = i + j + k$ $\quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} i - \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$

$= \langle 1,1,1 \rangle$ $\quad (1-0)i - (-1-0) + [0 - -1]$

$i + j + k$

$= \vec{n} \cdot (\vec{x} - \vec{p}) = 0$

$\vec{n} \cdot x = \vec{n} \cdot \vec{p}$

Kinda parametric form for the line of the intersection of the planes given by.

$x + y - z = 2$

must be on both planes ↓

$2x - y + 3z = 1$ ← how to set pos

$\vec{p} = \langle 1,1,0 \rangle$

$\vec{v} = \vec{n_1} \times \vec{n_2} = \langle 1,1,-1 \rangle \times \langle 2,-1,3 \rangle$

$= \langle 2, -5, -3 \rangle$

$f(t) = \langle 1+2t, 1-5t, -3t+0 \rangle$

Distances

Def

- the distance between sets is defined as the min of all distances between points in the respective sets

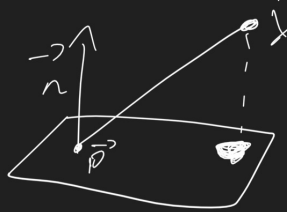
$$dist(X,Y) = \min_{x \in X,\, y \in Y} |x - y|$$

Point to plane:

find the distance from a pos $\vec{y}$ to a plane with normal $\vec{n}$ & pos $\vec{p}$

t = distance from pos $\vec{y}$ to the plane

$$\vec{n} \cdot (\vec{x} - \vec{p}) = c$$



$$\left| proj_{\vec{n}} (\vec{y} - \vec{p}) \right| = dist$$

Space Curves:

Defn:

A curve in $\mathbb{R}^n$ is the image of a vector valued function

$$r: \mathbb{R} \to \mathbb{R}^n$$

It has the form

$$r(t) = \langle x(t), y(t), z(t) \rangle$$

Input is scalar output vector.



ex: Twisted cubic

- $\langle t, t^2, t^3 \rangle$
- Helix: $\langle \cos(t), \sin(t), t \rangle$ (circle)

Find a Parameterization of the intersection of the cylinder $x^2+y^2=4$ & the plane $x-2y+4z=2$

we can take the curve

$$r(t) = \langle 2\cos(t), 2\sin(t), 0 \rangle$$

which gets us the circle

Calculus:

$$\lim_{t \to a} \vec{r}(t) = L$$

$$\lim_{t \to a} r(t) = \begin{bmatrix} \lim_{t \to a} x(t) \\ \lim_{t \to a} y(t) \\ \lim_{t \to a} z(t) \end{bmatrix}$$

$$r(t) = \langle \sin t, e^t - 1, \cos(t) - 1 \rangle$$

Compute the $\lim_{t \to c} \frac{1}{t} r(t)$ if it exists